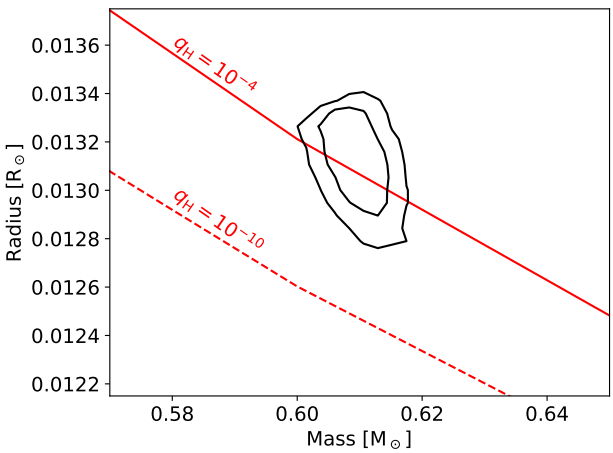

Figure 5. Constraints on the mass and radius of the white dwarf in ZTF J1828+2308 shown as contours (68 and 95 percentile regions). Also shown are theoretical mass-radius relationships for CO core white dwarfs from Bédard et al. (2020) with thick hydrogen envelopes ($q_H = 10^{-4}$, solid line) and with thin hydrogen envelopes ($q_H = 10^{-10}$, dashed line). A very good agreement is found with the thick envelope models.

that the companion to the white dwarf in ZTF J1828+2308 is a $0.0186 \pm 0.0008\,M_\odot$ ($19.5 \pm 0.8\,M_{Jup}$) object. This makes the companion the second lowest mass object transiting a white dwarf after the planet candidate around WD 1856+534 (excluding the transiting debris systems), although with a mass high enough that it is likely a brown dwarf rather than a planet.

We also find that the white dwarf has a mass of $0.610 \pm 0.004\,M_\odot$ and a radius of $0.0131 \pm 0.0002\,R_\odot$ and is therefore likely a CO core white dwarf (Figure 5) that reached the asymptotic giant branch (AGB), in contrast to many known close white dwarf plus brown dwarf binaries that contain low-mass He core white dwarfs (see the list in Zorotovic & Schreiber 2022 for examples). Figure 5 shows our mass and radius constraints relative to theoretical models for CO core white dwarfs with both thick and thin hydrogen envelopes (Bédard et al. 2020). Our measurements are in excellent agreement with the thick envelope models, as is generally found for white dwarfs in close binaries (Parsons et al. 2017a).

As a final step, we tested the effects of our choice of $f = 0.5$ as the emission-line correction factor. Due to the low values of the mass ratio ($q = 0.03$) and the scaled radius of the companion ($R_{BD}/a = 0.12$) the choice of f has only a minor effect on the final results (see Equation 1). This is because the companion is far from the centre of mass of the binary and small compared to the separation of the stars, so there is little difference in the velocity of its substellar point compared to its centre of mass. However, in order to see whether using the exact value of $f = 0.5$ has any effect we repeated our fit of the eclipse light curves, but this time forced the white dwarf to follow the mass-radius relationship of Bédard et al. (2020) but allowed f to vary freely (between 0 and 1). In this case we found a best fit value of $f = 0.52 \pm 0.02$ and all other parameters consistent with the previous fit. A value close to 0.5 is unsurprising, given the excellent agreement in our original fit between the white dwarf parameters and theoretical models in Figure 5. We are therefore confident that the values for the physical parameters of the brown dwarf are robust against the choice of f.

3.2 ZTF J123016.59−265551.34

ZTF J1230−2655 was discovered as an eclipsing white dwarf binary using data from ZTF (van Roestel et al. in prep). However, given its very southerly declination, the ZTF data coverage is fairly sparse and we made use of data from the Asteroid Terrestrial-impact Last Alert System (ATLAS, Tonry et al. 2018) to improve the ephemeris measurements and confirm the eclipsing nature of the system. Given that ZTF J1230−2655 sits on the white dwarf cooling track in the Gaia DR3 colour-magnitude diagram the companion cannot contribute a significant amount of the optical light.

The companion to the white dwarf in ZTF J1230−2655 is undetected in our follow up ULTRACAM light curves and we place a limit of $i_s > 23.2$ based on all of our in-eclipse observations. Using the Gaia DR3 parallax (5.11 ± 0.25 mas, Gaia Collaboration et al. 2023) this corresponds to $M_i > 16.7$, implying a spectral type of later than L2 (Kiman et al. 2019).

3.2.1 *Radial velocities*

Our X-shooter data of ZTF J1230−2655 do not reveal any clear evidence for irradiation-induced emission lines from the companion star at Hα. Moreover, while the narrow absorption core of the Hα line from the white dwarf is clear (see the left-hand panel of Figure 6), there are no obvious velocity variations. We attempted to use the same fitting procedure that we used for the spectra of ZTF J1828+2308 (removing the emission component), but failed to measure any significant velocity variations when fitting all the data simultaneously. We therefore opted instead to fit each spectrum individually with a combination of a first order polynomial and two Gaussian components (a wide component for the wings and a narrower one for the line core), forcing the two Gaussian components to have the same velocity.

The velocity measurements for the white dwarf in ZTF J1230−2655 are shown in the right-hand panel of Figure 6. The best fit sinusoid is also plotted, but the fit is poor and the data appear to just randomly scatter, rather than vary sinusoidally (although we unfortunately did not cover orbital phases near 0.75). In order to see if any sinusoidal signal is present in these data we performed two fits to the velocities, one in which we fitted a sinusoid to the data and another where we fitted the velocities as a constant value. In each case we also included an additional systematic error term in the fit and determined the model likelihood via

$$\ln \mathcal{L} = -\sum_i \left[\frac{(x_i - M_i)^2}{2(e_i^2 + \sigma^2)} + \ln \sqrt{2\pi(e_i^2 + \sigma^2)} \right], \quad (3)$$

where M_i is the predicted velocity (based on either the sinusoidal or constant term models), x_i and e_i are the velocity and error values for each measurement i, and σ is a systematic uncertainty term added to the measured uncertainties in quadrature. In both cases the fit gave a value for σ of around $5\,\mathrm{km\,s^{-1}}$, with the sinusoidal fit also having a semi-amplitude of $5\,\mathrm{km\,s^{-1}}$. We then performed a likelihood ratio test and found that we could not reject the constant fit and hence we have not detected any clear velocity variations in this system.

We placed an upper limit on K_{WD} by artificially injecting sinusoidal signals into our data with increasing amplitudes. We then re-ran the same fit and likelihood ratio test (with a fixed value of $\sigma = 5\,\mathrm{km\,s^{-1}}$) until we could reject the constant model. We found that we should have detected any signal with a semi-amplitude of $\geq 9\,\mathrm{km\,s^{-1}}$ and therefore we set an upper limit for the radial velocity semi-amplitude of the white dwarf in ZTF J1230−2655 of

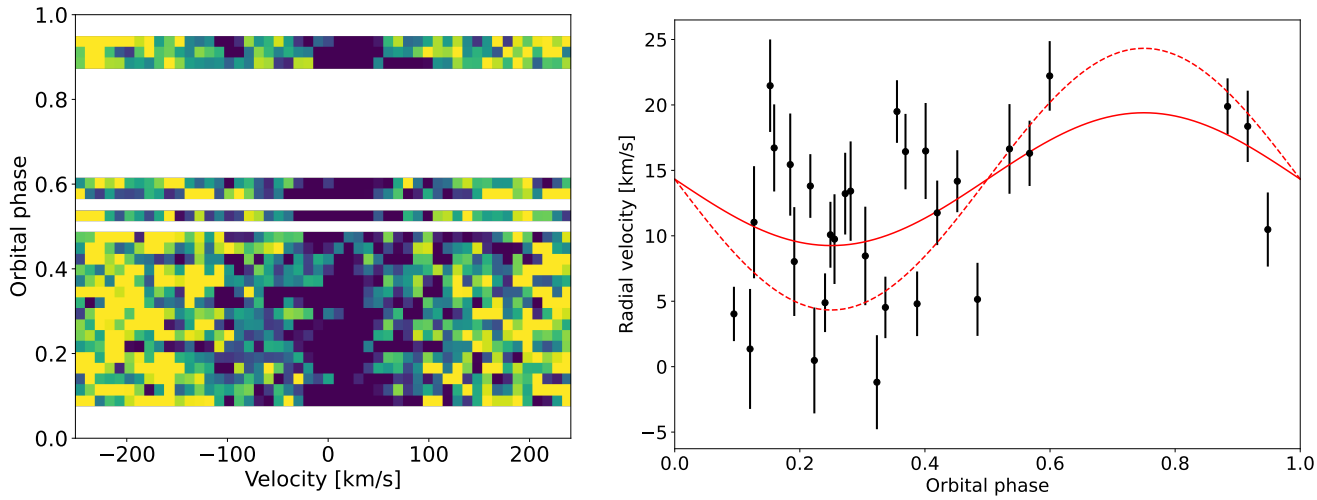

Figure 6. *Left:* phase-folded trailed spectra of the Hα line in ZTF J1230−2655. The non-LTE core of the white dwarf absorption line is clear, but there is no obvious velocity variations. *Right:* radial velocity measurements from the Hα absorption line along with a best fit (solid-red line), which is not statistically significant. Accounting for systematic uncertainties in these measurements of 5 km s^{-1} (not included in the plotted errorbars) we determine an upper limit on the radial velocity amplitude of 9 km s^{-1}, shown as the dashed-red line.

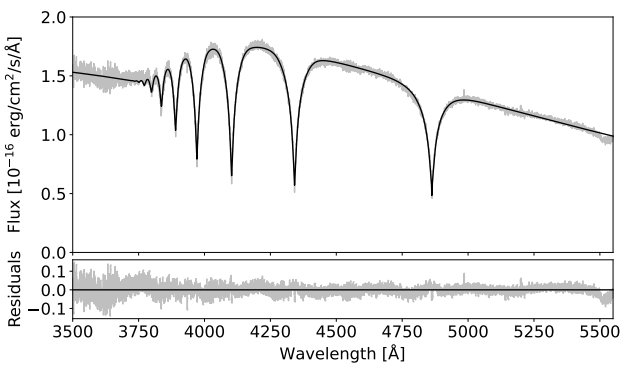

Figure 7. Averaged X-shooter UVB arm spectrum of ZTF J1230−2655, corrected only for the systemic velocity of the white dwarf (since no radial velocity variations were detected). The best fit Koester (2010) model is overplotted in black and the residuals to the fit are shown in the lower panel.

$K_{\rm WD} < 9\,{\rm km\,s^{-1}}$, which is shown as the dashed line in the right-hand panel of Figure 6.

3.2.2 *White dwarf spectral fit*

We averaged together all our X-shooter spectra of ZTF J1230−2655 in order to fit the white dwarf spectrum. Since we did not detect the motion of the white dwarf, we made no correction for this when averaging. We then fitted this spectrum with Koester (2010) models using the same method as described for ZTF J1828+2308. The average UVB arm spectrum and best fit model are shown in Figure 7. After applying the 3D corrections of Tremblay et al. (2013), which have a significant effect on the surface gravity, we found that the white dwarf in ZTF J1230−2655 has $T_{\rm eff} = 10000 \pm 110$ K and $\log(g) = 8.07 \pm 0.05$, where again we have applied the systematic uncertainties based on Hollands et al. (2024). This corresponds to a mass of around 0.65 M$_\odot$ based on the mass-radius relationship of Bédard et al. (2020).

As with ZTF J1828+2308, we also detect a weak Ca II 3934 Å absorption feature. Although we cannot rule out that this line moves with the white dwarf (since we don't detect the white dwarf's motion), the narrowness of the line leads us to conclude that this is likely an interstellar absorption feature. No other lines except the hydrogen Balmer series are detected in the X-shooter spectrum of ZTF J1230−2655.

3.2.3 *Light curve fit*

As with ZTF J1828+2308, our high-speed multi-band ULTRACAM eclipses of ZTF J1230−2655 can be used to place strong constraints on the stellar and binary parameters. However, in this case we have no direct measurement of $K_{\rm WD}$, just an upper limit, and no constraints on $K_{\rm BD}$ at all, since we did not detect any irradiation-driven emission lines in the spectra of ZTF J1230−2655. Therefore, a full model-independent fit is not possible in this case and we must rely on the use of a theoretical mass-radius relationship for the white dwarf in order to break the degeneracy between the two radii and orbital inclination. Even with the use of a mass-radius relationship, the fact that we only have an upper limit on $K_{\rm WD}$ means that we can only place an upper limit on the mass of the companion to the white dwarf.

The fit was performed in the same way as for ZTF J1828+2308, using the LCURVE code, the only differences being that we removed $R_{\rm WD}$ as an input and instead computed this value from the mass, temperature and Bédard et al. (2020) mass-radius relationship. We used the spectroscopically determined values of the white dwarf effective temperature and surface gravity as Gaussian priors. We also calculated $K_{\rm WD}$ for each model using Equation 2 and placed a uniform prior on this value with an upper limit of 9 km s^{-1}, to keep it consistent with our analysis of the radial velocity data. The combination of these constraints meant that we could precisely measure the orbital inclination and radius of the companion, while the white

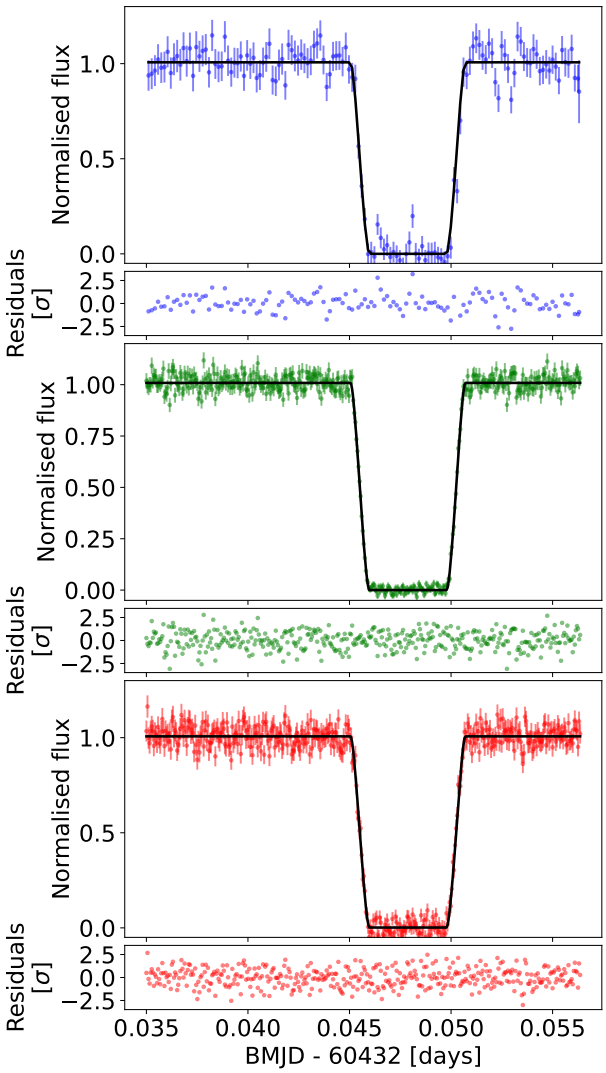

Figure 8. NTT+ULTRACAM eclipse light curves of ZTF J1230−2655 in the u_s (top), g_s (centre) and i_s (bottom) bands along with the best fit models (black lines). Residuals to the fits are shown beneath each panel (in standard deviations).

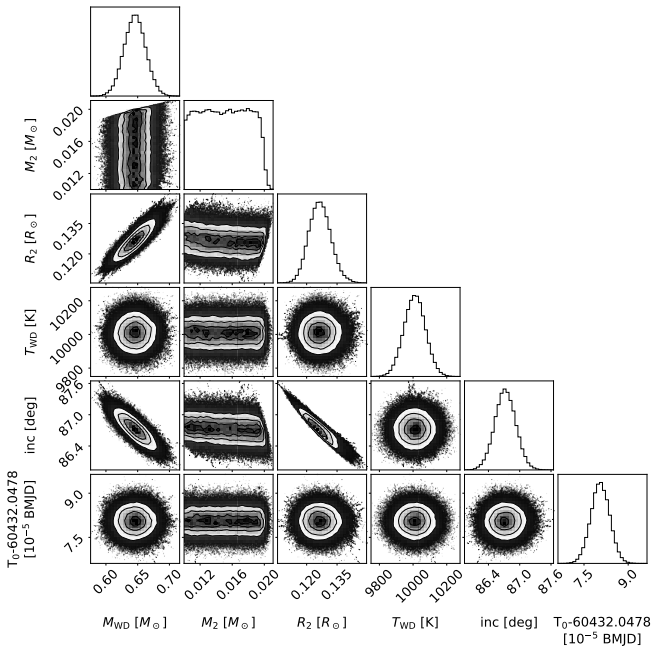

Figure 9. Posterior probability distributions for model parameters obtained through fitting the NTT+ULTRACAM light curves of ZTF J1230−2655.

dwarf parameters are determined by the spectroscopic fit, and the mass of the companion is constrained by Equation 2 and the upper limit on $K_{\rm WD}$. We performed an MCMC fit of the light curves in all three bands simultaneously using EMCEE using 100 walkers, with a burn-in period of 2000 and 10000 production steps.

The ULTRACAM eclipse light curves of ZTF J1230−2655 along with the best fit model are shown in Figure 8, the best fit parameters are listed in Table 2 and the MCMC parameter distributions are shown in Figure 9. We find that the companion to the white dwarf in ZTF J1230−2655 has a mass of $< 0.0211\,{\rm M}_\odot$ ($22.1\,{\rm M}_{\rm Jup}$), putting it in the same mass range as the companion to ZTF J1828+2308 and the transiting planet candidate around WD 1856+534. Indeed, with the present dataset we cannot rule out the possibility that this object has a planetary mass, although for the remainder of this paper we will refer to it as a brown dwarf. Furthermore, just like ZTF J1828+2308, the mass of the white dwarf in ZTF J1230−2655 ($0.646 \pm 0.017\,{\rm M}_\odot$) implies that the progenitor of the white dwarf evolved all the way to the AGB before the common envelope phase occurred.

4 DISCUSSION

4.1 Brown dwarf masses and radii

We have measured the mass and radius of the brown dwarf in ZTF J1828+2308 independent of any evolutionary models, while the radius and upper mass limit for the brown dwarf in ZTF J1230−2655 are only dependent on the (well tested) white dwarf mass-radius relationship (Parsons et al. 2017a). Therefore, these two objects are ideal for testing the mass-radius relationship at substellar masses. Figure 10 shows the measured masses and radii of transiting brown dwarfs, where the measurements are independent of evolutionary models. Also shown are a number of evolutionary models for different ages from Marley et al. (2021). Our two systems are highlighted in bold and likely have total ages around 1.5-2 Gyr (see Section 4.2). The measured mass and radius of the brown dwarf in ZTF J1828+2308 show good agreement with the evolutionary models, with no evidence of over-inflation, despite this object being moderately irradiated. However, despite not having an exact mass value for the brown dwarf in ZTF J1230−2655, the measured radius is clearly significantly larger than evolutionary models would predict, by around 20 per cent, regardless of the actual mass (if this is a planetary mass object it would still be around 10 per cent oversized). This is despite the fact that this object suffers very little irradiation from the cool white dwarf, and with an orbital period of 5.7 hours, the binary is well detached (only filling 59 per cent of its Roche lobe).

Over-inflated brown dwarfs in close binaries with white dwarfs have been found before. The brown dwarf in the eclipsing system WD 1032+011 also appears to be over-inflated (Casewell et al. 2020a; French et al. 2024), by possibly as much as 30 per cent, given the very